

22POULTRY

Product Requirements Document

Problem Definition & Opportunity Analysis

Phase 1: Market Research & User Discovery

Status: Research Complete | Draft V1.0

Document Date: May 11, 2026

Executive Summary

This document presents comprehensive market research and problem validation for 22POULTRY, a platform designed to transform India's small-scale poultry industry. Based on 2024-2026 market data and validated farmer economics, we have identified a critical market gap: small-scale poultry farmers (managing 1,000-10,000 birds) lose 45-55% of their potential revenue to middlemen while institutional buyers (QSRs, restaurants, hotels) struggle to source consistent, traceable, quality poultry from fragmented small farms.

Key Findings:

- **Market Size:** India's poultry market valued at ₹2.6 trillion (2025), growing at 13.8% CAGR to ₹8.4 trillion by 2034
- **Target User Base:** 6 million+ people employed in poultry farming, with 70% being small-scale farmers
- **Core Problem Validated:** Farmers receive only 45-55% of mandi price when selling through traditional middlemen
- **Buyer Demand Confirmed:** Food service market growing from \$51B (2024) to \$124B (2033) at 10.3% CAGR, with QSRs desperate for consistent B2B poultry sourcing
- **Competitive Landscape:** C2C agritech marketplaces have failed (funding dropped 95% from 2021 to 2024), but B2B institutional supply remains underserved

This document deliberately excludes solution exploration to maintain focus on problem validation and opportunity sizing. Solution design will follow in Phase 2 after stakeholder alignment on problem definition.

Document Purpose & Scope

Purpose

This PRD serves as the foundational research document for 22POULTRY's product development. It establishes:

- Problem definition with quantitative evidence
- Market opportunity sizing and growth projections
- Target user personas and their Jobs-to-be-Done (JTBD)
- Competitive landscape and market gaps
- User stories and acceptance criteria for validation

What This Document IS NOT

This document deliberately excludes:

- Solution design or feature specifications
- Technical architecture or implementation details
- UI/UX mockups or wireframes
- Go-to-market strategy or pricing models

These elements will be addressed in subsequent phases after stakeholder alignment on the problem definition.

Research Methodology

Data sources and validation:

- Market reports from IMARC, Expert Market Research, Mordor Intelligence (2024-2026)
- Government data: Department of Animal Husbandry & Dairying, NABARD schemes
- Competitive intelligence: Funding data from Venture Intelligence, Inc42 reports

- Academic research: Agricultural economics papers on middleman margins
- Industry reports: QSR procurement patterns, food service market analysis

Problem Statement

Core Problem Definition

Small-scale poultry farmers in India (managing 1,000-10,000 birds) face structural isolation that results in 25-40% lower income compared to their potential. This isolation manifests in four critical dimensions:

1. Market Access Isolation

Problem: Farmers lack direct access to institutional buyers and must sell through middlemen who extract 45-55% of the final selling price.

Evidence:

- Research from agricultural economics studies shows farmers receive only 45-55% of mandi price on average, with the margin between mandi price and farmer price larger in rural districts (65%) than in peri-urban areas (47%)
- Traditional middlemen provide essential services (credit, market knowledge, logistics) that farmers cannot access independently, creating dependency
- For a farmer selling 2,000 broiler birds at ₹200/bird wholesale price (₹4 lakh gross), the middleman margin reduces farmer revenue to ₹1.8-2.2 lakh — a loss of ₹1.8-2.2 lakh per cycle

2. Information Asymmetry

Problem: Farmers lack real-time market price transparency, quality standards knowledge, and awareness of institutional buyer requirements.

Evidence:

- Tracked prices by farmers are significantly closer to the prices they receive (₹2.15/kg) than to actual mandi prices (₹4.77/kg), indicating limited awareness of true market rates
- Mandi price variation is extreme year-over-year, with 87% of variation unexplained by seasonal or market-specific factors, making it impossible for farmers to forecast prices accurately

- QSRs and institutional buyers require traceability, food safety certifications, and consistent sizing/quality — standards that small farmers don't know how to meet

3. Capital Access Barriers

Problem: Small farmers lack access to affordable working capital for inputs (feed, chicks, medicine) and must rely on high-interest informal credit or supplier advances that lock them into unfavorable supply relationships.

Evidence:

- Feed costs account for 70% of total poultry farming expenses, creating massive working capital needs
- Formal loans for poultry farming (₹5-9 lakh typical farm setup) carry interest rates ranging from 7% (NABARD subsidized) to 26% per annum depending on lender, but many farmers cannot access formal credit
- When farmers take crop advances from commission agents or traders, they become obligated to sell to the same middleman at pre-agreed prices, reducing bargaining power

4. Network & Knowledge Isolation

Problem: Farmers lack peer networks for best practice sharing, quality training on disease management, and access to technical expertise that could improve yields and reduce mortality.

Evidence:

- Contract farming models in Tamil Nadu and Maharashtra provide training and technical support to farmers, demonstrating that knowledge transfer improves productivity
- Government extension services exist but are under-resourced, with large integrators providing superior training to their contract farmers

- Mortality rates and disease outbreaks (avian influenza, Newcastle disease) disproportionately affect small farmers who lack biosecurity knowledge and vaccination protocols

Why Existing Solutions Have Failed

The agritech marketplace model collapse (2021-2024) provides critical insights into what NOT to build:

C2C Marketplace Failure

Data: Agritech marketplaces raised \$490 million in 2021 but saw funding collapse to just \$26 million in 2024 (95% decline). Major players like WayCool struggled with massive losses despite high valuations.

Why they failed:

- Farmers remain skeptical of digital platforms, preferring traditional middlemen who understand local agricultural cycles and provide crucial credit support
- High operational costs from cold chain, logistics, and farmer acquisition made unit economics unsustainable
- Consumer demand for direct farm-to-table purchasing was overestimated; traditional retail remains dominant (45% market share in 2025)
- Platforms tried to eliminate middlemen entirely without providing equivalent value (credit, local knowledge, relationship-based trust)

What IS Working: Contract Farming Model

Data: Contract poultry farming is an emerging trend particularly in Tamil Nadu and Maharashtra, supported by government subsidies. The National Livestock Mission provides up to 50% of project cost (max ₹25 lakh) for farm establishment. This model empowers small farmers through input support, assured B2B retention, and training.

Why it works:

- Guaranteed buyer (integrator) provides certainty that farmers value over spot market volatility
- Upfront input support (feed, chicks, technical advice) solves the working capital problem

- Training and quality standards from integrators improve farmer yields and reduce mortality
- Relationship-based model aligns with farmer preferences for trusted partners over anonymous digital platforms

Problem Impact: Quantified Farmer Economics

To validate problem severity, we calculated the financial impact on a typical small-scale broiler farmer:

Metric	Current (via Middleman)	Potential (Direct B2B)	Farmer Gain
Farm size	2,000 birds per cycle	2,000 birds per cycle	—
Price per bird	₹200 (wholesale)	₹220-240 (direct)	+₹20-40/bird
Gross revenue per cycle	₹4.0 lakh	₹4.4-4.8 lakh	+₹40K-80K
Farmer revenue (after middleman cut)	₹1.8-2.2 lakh (45-55%)	₹4.2-4.6 lakh (95-98%)	+₹2.0-2.8 lakh
Net profit per cycle	₹60-80K	₹75-105K	+25-30% increase
Annual income (8 cycles/year)	₹4.8-6.4 lakh	₹6.0-8.4 lakh	+₹1.2-2.0 lakh/year

Source: Broiler profit margins from agricultural economics research (₹25-40 per bird), middleman margin data from field studies showing farmers receive 45-55% of mandi price.

Critical finding: A 25-30% income increase (₹1.2-2.0 lakh per year) is transformative for a small-scale farmer household. This represents the difference between subsistence farming and the ability to invest in farm expansion, children's education, or improved living standards.

Market Opportunity

Market Size & Growth

Market Segment	2025 Value	2033-2035 Projection
India Poultry Market	₹2.6 trillion (\$32.9B)	₹8.4 trillion (\$71.8B) 13.8% CAGR
Food Service Market	\$51.0B	\$123.5B (2033) 10.33% CAGR
Poultry Feed Market	₹955B	₹1,291B (2033) 3.23% CAGR
Agritech Market (India)	\$9B	\$28B (2030) 25% CAGR

Sources: IMARC Group, Expert Market Research, Mordor Intelligence, Inc42 Agritech Market Landscape Report 2025

Supply-Side Opportunity

Target Farmer Segment

- Total addressable market: 6+ million people employed in poultry farming in India
- Small-scale farmers: 70% of contract broiler farmers manage flocks of 3,000-10,000 birds, making this the dominant segment
- Geographic concentration: Maharashtra, Tamil Nadu, Andhra Pradesh, Telangana, West Bengal, Karnataka account for majority of production
- Production scale: India produces 140+ billion eggs and 4.5 million tonnes of broiler meat annually (2024)

Demand-Side Opportunity: B2B Institutional Buyers

- QSR/restaurant sector: Intensive growth of organized food service restaurants creating great incremental demand for poultry, with chicken as a widely used staple ingredient
- B2B procurement acceleration: Institutional buyers, QSRs, and cloud kitchens signing agreements for hygienic, portioned, and traceable poultry products
- Digital procurement: Startups like Captain Fresh digitizing procurement with real-time inventory, pricing, and fulfillment solutions that appeal to enterprise clients seeking predictability
- Cold chain investment: Digitally integrated cold chain logistics allowing real-time temperature monitoring, reducing spoilage, enhancing safety, and boosting trust among institutional buyers

Market Gap: The Missing Link

Current situation: QSRs and institutional buyers source from large integrators (Suguna, Venky's, IB Group) or scattered informal suppliers. Small-scale farmers cannot compete because they lack aggregation, traceability systems, and logistics coordination.

The opportunity: No platform exists to efficiently connect small-scale farmers directly to institutional buyers with:

- Quality standardization and traceability
- Aggregated consistent supply from fragmented small farms
- Logistics coordination for 3x/week pickups
- Credit/input support that aligns farmer and buyer incentives

Target Users & Personas

Primary User Persona 1: Small-Scale Broiler Farmer

Attribute	Details
Name	Rajesh Kumar (representative persona)
Age	35-50 years old
Location	Rural/peri-urban Maharashtra, Tamil Nadu, Andhra Pradesh, or Karnataka
Farm Size	2,000-5,000 broiler birds per 6-week cycle
Education	8th-12th grade; semi-literate; basic smartphone literacy
Tech Access	Smartphone (Android); WhatsApp user; 3G/4G connectivity (intermittent)
Annual Income	₹4.8-6.4 lakh from poultry (current via middlemen)
Primary Language	Hindi, Marathi, Tamil, Telugu, or Kannada (regional variation)

Pain Points

- "I have no choice but to sell to the local middleman. He gives me ₹180-200 per bird, but I know he sells for ₹350-400. I lose ₹1.5-2 lakh every cycle."*
- "Feed costs keep rising. I spend ₹2.8 lakh on feed alone per cycle but don't know if I'm getting the best price or quality."*
- "I need credit to buy chicks and feed upfront, but the middleman gives me advances at 18-24% interest. Then I'm stuck selling to him at whatever price he decides."*
- "I lost 15% of my last batch to disease. No one taught me proper vaccination schedules or biosecurity. The big integrators train their contract farmers but I'm on my own."*

- *"I hear hotels and restaurants need chicken but I have no way to reach them. I don't have cold storage or a truck. How can I compete with the big companies?"*

Jobs to Be Done (JTBD)

- **Primary JTBD: Sell my birds at fair market price (70-80% of retail, not 45-55%) to a guaranteed buyer**
- Secondary JTBD: Access affordable working capital for inputs without predatory interest or lock-in
- Tertiary JTBD: Learn best practices to reduce mortality and improve yields

Primary User Persona 2: QSR/Restaurant Procurement Manager

Attribute	Details
Name	Priya Sharma (representative persona)
Role	Operations/Procurement Manager at multi-location QSR or cloud kitchen chain
Restaurant Type	QSR chain (5-20 outlets), cloud kitchen, or mid-scale casual dining (2-5 locations)
Location	Metro cities (Mumbai, Bangalore, Delhi, Hyderabad)
Monthly Poultry Spend	₹1.5-3 lakh per outlet (₹30-40K per week)
Education	College degree; tech-savvy; comfortable with procurement software

Pain Points

- "We source from 3-4 different local suppliers. Quality is inconsistent - sometimes the chicken is undersized, sometimes delivery is late. We can't serve customers if we run out."*
- "Food safety is critical. We need traceability - farm ID, vaccination records, slaughter date. Most small suppliers can't provide this documentation."*
- "The big integrators (Suguna, Venky's) are reliable but expensive. They charge premium prices because we're locked in. I wish there was a way to source directly from farms at better rates."*
- "Coordinating with 10+ small suppliers is a nightmare. Different payment terms, different delivery schedules, constant phone calls. I need one platform that aggregates supply."*

Jobs to Be Done (JTBD)

- Primary JTBD: Get consistent, traceable, quality poultry at competitive prices with 3x/week guaranteed delivery**

- Secondary JTBD: Reduce procurement complexity by consolidating multiple suppliers into one reliable partner
- Tertiary JTBD: Meet food safety compliance requirements with full traceability documentation

User Stories & Acceptance Criteria

The following user stories define the core jobs-to-be-done without prescribing specific solutions. Acceptance criteria establish measurable validation gates.

Epic 1: Farmer Market Access

User Story 1.1: Direct Buyer Access

As a **small-scale broiler farmer managing 2,000-5,000 birds,**

I want to **sell my birds directly to institutional buyers (restaurants, hotels, QSRs),**

So that **I can earn 70-80% of the retail price instead of 45-55% through middlemen, increasing my profit by 25-30% per cycle.**

Acceptance Criteria:

- Given I have 2,000 birds ready for sale, When I connect with a buyer, Then I receive a price quote that is at least 10% higher than my current middleman rate
- Given I complete a sale, When payment is processed, Then I receive funds within 48 hours of delivery (vs. 7-15 days from middlemen)
- Given I make a sale through direct buyer channel, When I calculate my net income, Then I earn ₹15,000-25,000 more per cycle than selling via traditional middleman

User Story 1.2: Price Transparency

As a **small-scale poultry farmer with limited market knowledge,**

I want to **see real-time market prices for poultry in my region,**

So that **I can negotiate better prices and know whether I'm getting a fair deal.**

Acceptance Criteria:

- Given I open the platform, When I view market prices, Then I see current live bird prices from at least 3 different mandi/market sources in my district
- Given I receive a price offer, When I compare it to market rate, Then the offer is within 5% of the published mandi price (preventing exploitation)
- Given market prices fluctuate, When prices change by more than 10%, Then I receive a notification via WhatsApp or SMS so I can time my sales better

User Story 1.3: Guaranteed Buyer Contracts

As a farmer who invests ₹2.8 lakh in feed and inputs upfront,

I want to have a guaranteed buyer committed to purchasing my birds before I start a new batch,

So that I don't take on market risk and can plan my cash flow with certainty.

Acceptance Criteria:

- Given I have a buyer interested in my supply, When I sign an agreement, Then the buyer commits to purchasing a specific quantity (e.g., 1,500-2,000 birds) at a pre-agreed price floor
- Given my birds are ready for sale, When I notify the buyer, Then pickup is scheduled within 48 hours (no more than 3 days delay)
- Given the buyer commits to purchase, When the pickup happens, Then the buyer accepts at least 95% of my birds (rejections <5% due to quality issues)

Epic 2: QSR/Buyer Supply Reliability

User Story 2.1: Consistent Quality Supply

As a QSR operations manager needing 150 kg of chicken per week across 5 outlets,

I want to receive consistent, sized, fresh poultry from a reliable aggregated source,

So that my menu quality remains consistent and I never run out of chicken during service hours.

Acceptance Criteria:

- Given I place a weekly order for 150 kg, When delivery arrives, Then 95%+ of birds meet my weight specification (1.5-2.0 kg live weight)
- Given I have scheduled 3x/week deliveries (Mon/Wed/Fri), When delivery day comes, Then my order arrives within the 2-hour scheduled window at least 90% of the time
- Given I receive a shipment, When I inspect quality, Then rejection rate is <5% (spoilage, size issues, disease visible)

User Story 2.2: Food Safety & Traceability

As a restaurant owner subject to FSSAI inspections and food safety audits,

I want to have complete traceability records for all poultry I purchase (farm ID, vaccination records, slaughter date),

So that I can comply with food safety regulations and protect my customers from foodborne illness.

Acceptance Criteria:

- Given I receive a poultry shipment, When I scan/access the batch ID, Then I can see: farm location, farmer ID, bird age, vaccination dates, feed type, and slaughter date

- Given an FSSAI inspector requests documentation, When I provide traceability records, Then all required fields are present and verifiable within 5 minutes
- Given a food safety incident (customer complaint), When I need to trace the source, Then I can identify the exact farm and batch within 15 minutes for recall purposes

User Story 2.3: Cost Competitiveness

As a **cloud kitchen operator managing tight margins (8-12% net profit)**,

I want to **source poultry at 10-15% lower cost than large integrators**,

So that I can **maintain competitive menu pricing while improving my gross margins**.

Acceptance Criteria:

- Given I receive price quotes from both platform and integrator (Suguna/Venky's), When I compare rates, Then the platform price is 10-15% lower for equivalent quality and service
- Given I purchase 150 kg/week, When I switch from integrator to platform, Then I save ₹3,000-5,000 per month in poultry procurement costs
- Given I evaluate total cost of ownership (price + delivery fees + quality), When I calculate net cost, Then the platform delivers at least 8% total cost savings vs. status quo

Epic 3: Platform Trust & Reliability

User Story 3.1: Payment Reliability (Farmer Trust)

As a farmer who has been cheated by buyers in the past (delayed payments, quality disputes),

I want to receive guaranteed payment within 48 hours of delivery,

So that I can pay my feed suppliers and workers on time without cash flow stress.

Acceptance Criteria:

- Given I deliver birds to a buyer, When quality inspection is complete, Then payment is initiated automatically within 6 hours
- Given payment is initiated, When I check my bank account, Then funds are received within 48 hours for 95%+ of transactions
- Given a quality dispute arises, When resolution is needed, Then I receive at least 85% of the agreed payment even in worst-case scenarios (protection against total loss)

User Story 3.2: Support in Local Language

As a farmer who speaks Marathi/Tamil/Telugu and has limited English proficiency,

I want to access platform support and documentation in my native language,

So that I can understand contracts, get help when needed, and use the platform confidently.

Acceptance Criteria:

- Given I need customer support, When I contact the platform, Then I can communicate via WhatsApp or phone in my regional language (Hindi, Marathi, Tamil, Telugu, Kannada)
- Given I have a question, When I reach out for help, Then I receive a response from a local-language support agent within 4 hours during business hours

- Given I need to understand terms and conditions, When I view contracts, Then all critical documents are available in both English and my regional language

Success Metrics & Validation Gates

These metrics will validate whether the problem is being solved effectively. Metrics are defined before solution design to ensure we measure outcomes, not outputs.

Phase 1 Validation Metrics (Months 1-6)

Metric	Baseline	Target (Month 6)	Success Criteria
Farmer Income Improvement	45-55% of mandi price	70-80% of retail price	25-30% net income increase validated through surveys
Farmer Net Promoter Score	N/A (new platform)	NPS >40	Farmers actively recommend to peers
QSR Order Fulfillment Rate	N/A	>90% on-time delivery	Buyers trust platform as reliable supplier
Monthly GMV (Gross Merchandise Value)	₹0	₹5-10 crore	Proves market demand and unit economics
Active Farmers Onboarded	0	200-500 farmers	Supply aggregation proof in 1 pilot metro
Active QSR/Restaurant Buyers	0	50-100 buyers	Demand validation in 1 pilot metro
Buyer Repeat Order Rate	N/A	>80% weekly reorder	Proves product-market fit and retention

Critical Assumptions to Validate

These assumptions underpin the problem definition. If any prove false, we must revisit the strategy:

Supply-Side Assumptions (Farmer)

- **Assumption 1: Farmers will switch from traditional middlemen if offered 20-30% higher prices with guaranteed payment**
 - Validation method: Pilot with 20-30 farmers; measure switching rate and satisfaction
 - Invalidation signal: <50% of farmers approached agree to try platform
- **Assumption 2: Small-scale farmers can meet QSR quality and consistency requirements with minimal training**
 - Validation method: Run quality audits on first 50 farmer deliveries; measure rejection rate
 - Invalidation signal: >20% rejection rate by QSRs due to quality/sizing issues
- **Assumption 3: Farmers have basic smartphone literacy and can use WhatsApp-based interfaces**
 - Validation method: User testing with 10-15 farmers from target segment
 - Invalidation signal: >40% cannot complete basic tasks even with phone support

Demand-Side Assumptions (QSR/Buyer)

- **Assumption 4: QSRs/restaurants will buy from small-farm aggregators if traceability and consistency are guaranteed**
 - Validation method: Interview 15-20 QSR procurement managers; present value prop and gauge interest
 - Invalidation signal: <60% express willingness to trial the platform
- **Assumption 5: QSRs can save 10-15% on procurement costs vs. integrators while maintaining quality**
 - Validation method: Compare platform pricing to integrator rates across 10+ QSRs

- Invalidation signal: Net savings <5% after accounting for quality/logistics issues
- **Assumption 6: Logistics coordination (3x/week pickups from 200+ farms to 50+ restaurants) is operationally feasible**
 - Validation method: Pilot logistics in one metro for 2 months; measure fulfillment rate and cost
 - Invalidation signal: <85% on-time delivery or logistics cost >5% of GMV

Risks & Mitigation Strategies

Risk	Impact	Mitigation Strategy
Farmers prefer traditional middlemen relationships despite better pricing	High - kills supply side	Validate in pilot: if <50% switch rate, pivot to input supply or integrator partnerships
Small farmers cannot meet QSR quality standards consistently	High - kills demand side	Build quality training program; partner with existing integrators for quality protocols; accept only certified farmers
Integrators (Suguna, Venky's) undercut pricing or lock farmers into exclusive contracts	Medium - competitive response	Focus on farmers not in integrator contracts; differentiate on payment speed + transparency; explore partnership vs. competition
Logistics complexity makes 3x/week delivery uneconomical	High - breaks unit economics	Pilot logistics first; if cost >5% GMV, shift to buyer-arranged logistics or hub-and-spoke model
Regulatory barriers for direct farm-to-restaurant sales (FSSAI licensing)	Medium - solvable with process	Validate regulations in pilot city; work with FSSAI-approved slaughterhouses; build compliance into onboarding
Disease outbreak (avian flu) disrupts supply and demand simultaneously	High - industry-wide risk	Geographic diversification; maintain relationships with integrators as backup; insurance products for farmers

Next Steps & Decision Points

Immediate Action Items (Week 1)

- **Validate QSR demand: Conduct 5-10 phone interviews with restaurant procurement managers in pilot metro (Mumbai or Bangalore)**
 - Goal: Confirm willingness to source from small-farm aggregator
 - Success criteria: >60% express interest in trialing platform
- **Validate farmer supply: Visit 5-10 small-scale broiler farmers in peri-urban areas**
 - Goal: Confirm switching willingness and validate economics (middleman margins, current pricing)
 - Success criteria: >50% willing to try if offered 20-30% higher price
- **Regulatory check: Consult with municipal food safety officer or FSSAI representative**
 - Goal: Identify licensing requirements for direct farm-to-restaurant sales
 - Red flag: If licensing cost >₹50K per farmer or takes >3 months, may need integrator partnership model
- **Competitive intelligence: Map existing B2B poultry suppliers in pilot metro**
 - Goal: Understand competitive pricing, quality standards, and market gaps
 - Red flag: If integrators already offer direct small-farm sourcing at similar economics, may need differentiation pivot

Decision Gates

Week 2 Gate: Demand Validation

- Decision: Proceed to solution design if >60% of QSRs express interest

- Pivot: If <60%, explore alternative buyer segments (hotels, caterers, processed meat companies)

Week 2 Gate: Supply Validation

- Decision: Proceed if >50% of farmers willing to switch at proposed pricing
- Pivot: If <50%, consider input supply marketplace (feed procurement) or integrator partnership model

Week 3 Gate: Regulatory Feasibility

- Decision: Proceed if licensing process is <3 months and <₹50K per farmer
- Pivot: If regulatory barriers too high, partner with FSSAI-approved slaughterhouses or work through integrator licenses

Phase 2 Initiation Criteria

Proceed to solution design and MVP build ONLY if all three gates pass:

- ✓ Demand validated: >60% QSR interest confirmed
- ✓ Supply validated: >50% farmer switching willingness confirmed
- ✓ Regulatory path clear: Licensing feasible within acceptable time/cost

If ANY gate fails, revisit the problem definition and consider documented pivot options before investing in solution development.